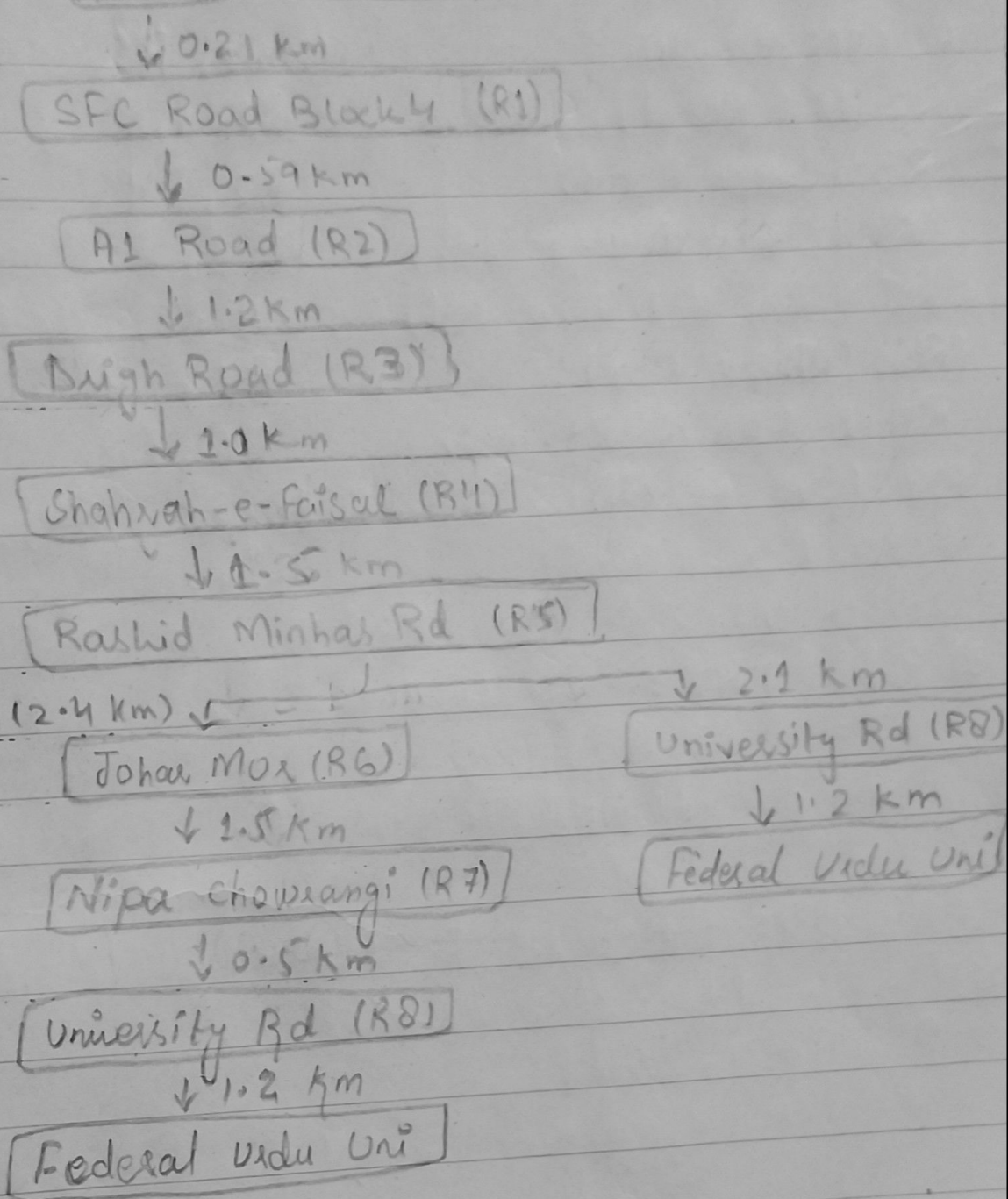


Route Map :-



Route 1 :-

$$0.2 + 0.59 + 1.2 + 1.0 + 1.5 + 2.4 + 1.5 + 0.5 + 1.2 = 10.09 \text{ km}$$

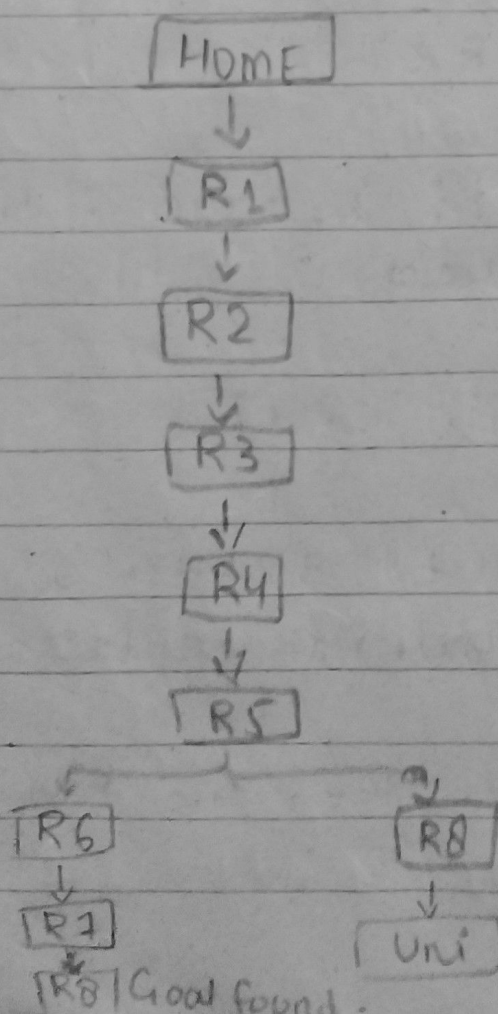
Route 2 :-

$$0.2 + 0.59 + 1.2 + 1.0 + 1.5 + 2.1 + 1.2 = 7.7 \text{ km}$$

2) Graph Representation :-

Node	Connected to
Home	R1
R1 (SFC blk. 4)	Home, R2
R2 (A1 Road)	R1, R3
R3 (Drigh Road)	R2, R4
R4 (Shahsah-e-faisal)	R3, R5
R5 (Rashid Minhas Rd)	R4, R6, R8 (bypass)
R6 (Johar Max)	R5, R7
R7 (Nipa Chowang)	R6, R8
R8 (Uni Road)	R5 (bypass), R7, Uni
Federal Urdu Uni	R8

3) BFS Tree:



Goal found.

4) BFS Trace :-

Step 0: Initializing.

$Q = \{ \text{Home} \}$

$V = \{ \text{Home} \}$

Step 1: Dequeue Home $\rightarrow$ Expand Home
 $\rightarrow R_1$ (Shah Faisal Block 4 Rd).

$Q = \{ R_1 \}$

$V = \{ \text{Home}, R_1 \}$

Step 2: Dequeue R_1 (Shah Faisal Blk 4 Rd)
 $\rightarrow$ Expand $R_1 \rightarrow R_2$ (A1 Road).

$Q = \{ R_2 \}$

$V = \{ \text{Home}, R_1, R_2 \}$

Step 3: Dequeue R_2 (A1 Road) $\rightarrow$ Expand
 $R_2 \rightarrow R_3$ (Digh Road).

$Q = \{ R_3 \}$

$V = \{ \text{Home}, R_1, R_2, R_3 \}$

Step 4: Dequeue R_3 (Digh Road) $\rightarrow$
Expand $R_3 \rightarrow R_4$ (Shah-e-faisal).

$Q = \{ R_4 \}$

$V = \{ \text{Home}, R_1, R_2, R_3, R_4 \}$.

Step 05: Dequeue R4 (Shahrah-e-Iqbal)
→ Expand R4 → R5 (Rashid Minhas Rd.)
 $Q = \{R5\}$
 $V = \{Home, R1, R2, R3, R4, R5\}$

Step 06: Dequeue R5 (Rashid Minhas Rd) → Expand R5 → R6 (Johar Mor), R8 (Uni Rd).

$Q = \{R6, R8\}$
 $V = \{Home, R1, R2, R3, R4, R5, R6, R8\}$

Step 07: Dequeue R6 (Johar Mor) → Expand R6 → R7 (Nipa Chowlangi).

$Q = \{R8, R7\}$
 $V = \{Home, R1, R2, R3, R4, R5, R6, R8, R7\}$

Step 08: Dequeue R8 (Uni Rd) → Expand R8 → Uni (Federal Urdu Uni).

$Q = \{R7, Uni\}$

$V = \{Home, R1, R2, R3, R4, R5, R6, R8, R7\}$

Step 09: Dequeue R7 (Nipa Chowlangi) → Expand R7 → R8 (Uni Road) → Already visited

$Q = \{Uni\}$

Step 10: Dequeue Uni (Federal Urdu Uni) → Goal Found.

Shortest Path: -

Home $\rightarrow$ R1 $\rightarrow$ R2 $\rightarrow$ R3 $\rightarrow$ R4 $\rightarrow$ R5 $\rightarrow$ R8 $\rightarrow$

Uni.